

Problem set #1

MATH 4503/6503: Numerical Methods for Differential Equations

Due date: January 31, 2026

Reading resources: Notes uploaded to D2L; book pages 3-11 and 247-248

Problem #1: big- & little-"oh" notation

Hey, everyone! Welcome to the first of your assignments! Let us start slow, simple, and easy. We have covered the big- and little-"oh" notation in class, and you are aware of the definitions. In this problem, I want you to use that knowledge to answer some simple questions in Maple.

You will have to make good use of the `limit` and `Limit` functions. The code should be available to you from the "Calculus" palette, and you could get all details on functionality from the relevant help page (remember commands like `?limit`, `?Limit`, etc.). Note that you are supposed to be taking right-hand limits ($h \rightarrow 0^+$) as (our step size) h cannot be negative.

- (a) If $f(h) = 5h \tan(h^3)$ and $g(h) = h^4$, is $f(h) = O(g(h))$? Is $f(h) = o(g(h))$? If $g(h) = \sqrt{h + h^2}$, is $f(h) = o(g(h))$? Is $f(h) = O(g(h))$ in this latter case?
- (b) Find explicit examples of functions $f(h)$ and $g(h)$ such that $f(h) = o(g(h))$ but $f(h) \neq O(g(h))$. If this is not possible, why?
- (c) Consider the polynomial $u(x) = 7 + x^2 + 4x^5 + x^{10}$. The following is a finite difference approximation to $u'''(x)$:

$$-\frac{2u(x-2h)}{5h^3} + \frac{5u(x-h)}{8h^3} + \frac{2u(x)}{5h^3} - \frac{7u(x+h)}{8h^3} + \frac{11u(x+3h)}{40h^3} - \frac{u(x+5h)}{40h^3}.$$

If the error in it is $O(h^p)$, what is the largest possible value of p ? (*hint: expand and collect commands will come in handy.*)

Problem #2: higher derivative approximations

One of the ways we learnt in class to find finite difference approximations to higher derivatives was to repeatedly apply approximations to first order derivatives. We play with this technique in this problem. You will need the d_p, d_m, d_c functions I defined in the Maple worksheet on finite

differencing as they implement the approximations

$$D_+u(x) = \frac{u(x+h) - u(x)}{h},$$

$$D_-u(x) = \frac{u(x) - u(x-h)}{h},$$

$$D_0u(x) = \frac{u(x+h) - u(x-h)}{2h},$$

respectively.

- Use the functions d_p, d_m, d_c to work out $D_+D_+D_+D_+u(x)$ and $D_-D_-D_-D_-u(x)$. Are these centered approximations?
- Also work out $D_+D_0D_-D_0u(x)$ and $D_0D_0D_-D_+u(x)$. Are these centered? What do you think determines if an approximation is centered in the context of repeatedly applying approximations to the first derivative?

Hint: Go in steps. For instance, if you want to find the expression for $D_+D_-u(x)$, use d_m to calculate $D_-u(x)$ in terms of $u(x)$ and $u(x-h)$ and then apply d_p to $u(x)$ and $u(x-h)$, combining the resulting expressions appropriately to get one for $D_+D_-u(x)$.

Problem #3: functions and procedures

- The previous problem must have given you some acquaintance with the idea of functions in Maple. Let us explore it a bit further in this problem. Define the following function in Maple:

$$f(\alpha) = \int_0^\infty dx \frac{x^\alpha}{1+e^x}.$$

Then, plot $f(\alpha)$ against α for $\alpha \in [0, 5]$.

- What we have just defined is called a mapping in Maple, or a function sometimes. What other programming languages refer to as functions Maple calls procedures. We will be writing a simple procedure in this part. The Lucas numbers are defined by

$$U_0 = 0, U_1 = 1, U_n = PU_{n-2} - QU_{n-1}, n \geq 2$$

for two parameters P and Q . Write a Maple procedure that takes a positive integer N and two numbers P and Q as its arguments and returns the numerical value of $S_N^{(P,Q)}$, where

$$S_N^{(P,Q)} = \sum_{n=0}^N U_n.$$

Your procedure should work for all $N \geq 1$.

Problem #4: differential equations

We finally get to actual differential equations: both ODEs and PDEs! The purpose of this problem is to give you experience with the Maple function `dsolve`.

- (a) In Maple, solve the following initial value problem and plot the solution for $x \in [-10, 10]$.

$$\psi'' + 5x^2\psi = 0, \psi(0) = 1, \psi'(0) = 0.$$

- (b) Electromagnetic waves propagating in a coaxial cable (i.e., for T.V. signals) are governed by a partial differential equation for a potential $\psi = \psi(t, \rho, \phi, z)$:

$$\frac{\partial^2 \psi}{\partial t^2} - \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \psi}{\partial \rho} \right) - \frac{1}{\rho^2} \frac{\partial^2 \psi}{\partial \phi^2} - \frac{\partial^2 \psi}{\partial z^2} = 0.$$

- (i) Substitute $\psi(t, \rho, \phi, z) = e^{im\phi} e^{ikz} e^{-i\omega t} f(\rho)$ into the PDE, and simplify to obtain an ODE for $f(\rho)$ (*hint: subs, simplify, and isolate commands might be helpful*).
- (ii) Find the general solution for f using `dsolve`.